

Docker-01

What is Docker?

—> **Docker is a containerization platform used to run applications inside lightweight containers.**

“Before Docker, applications were deployed using virtualization. Docker introduced containerization, which is faster and lightweight.”

Virtualization (Virtual Machine) – Key Points

1. Virtualization allows running **multiple operating systems** on a single physical machine.
2. Each virtual machine has its **own full operating system** and kernel.
3. A **hypervisor** (like VirtualBox, UTM, VMware) is used to create and manage virtual machines.
4. Virtual machines are **heavy** because they include a full OS.
5. Virtual machines take **more time to start** compared to containers.
6. Each VM is **fully isolated** from others.
7. Virtualization is used to **test OS, servers, and applications** safely.
8. Example: Creating an **Ubuntu VM** and installing Nginx.

Use case: When a full OS environment is required.

Docker (Containerization) – Key Points

1. Docker is a **containerization platform** used to run applications in containers.
2. Containers **do not need a full operating system**.
3. All containers **share the host OS kernel**.
4. Containers are **lightweight and fast**.

5. Docker packages the **application and its dependencies** together.
6. Containers start in **seconds**.
7. Docker ensures the application runs **the same in all environments**.
8. Docker uses **images** to create containers.
9. Docker reduces **resource usage** (RAM and CPU).
10. Docker is widely used in **DevOps and cloud environments**.

Use case: When fast, scalable, and portable applications are needed.

Virtualization vs Docker

Virtualization	Docker
Full OS	No full OS
Heavy	Lightweight
Slow startup	Fast startup
Uses hypervisor	Uses Docker engine
More RAM usage	Less RAM usage

Tasks :

Task :1

1. Setup a virtual machine using Ubuntu image.

Step 1: Download Requirements

1. Download Ubuntu ISO

	WhatsApp Image 2025-10-16 at 14.27.26.jpeg	238 KB	JPEG image
	ubuntu-24.04.3-live-server-arm64.iso	3 GB	ISO Disk Image
	Docker.dmg	546.9 MB	Disk Image
	TeamViewer.dmg	133.6 MB	Disk Image

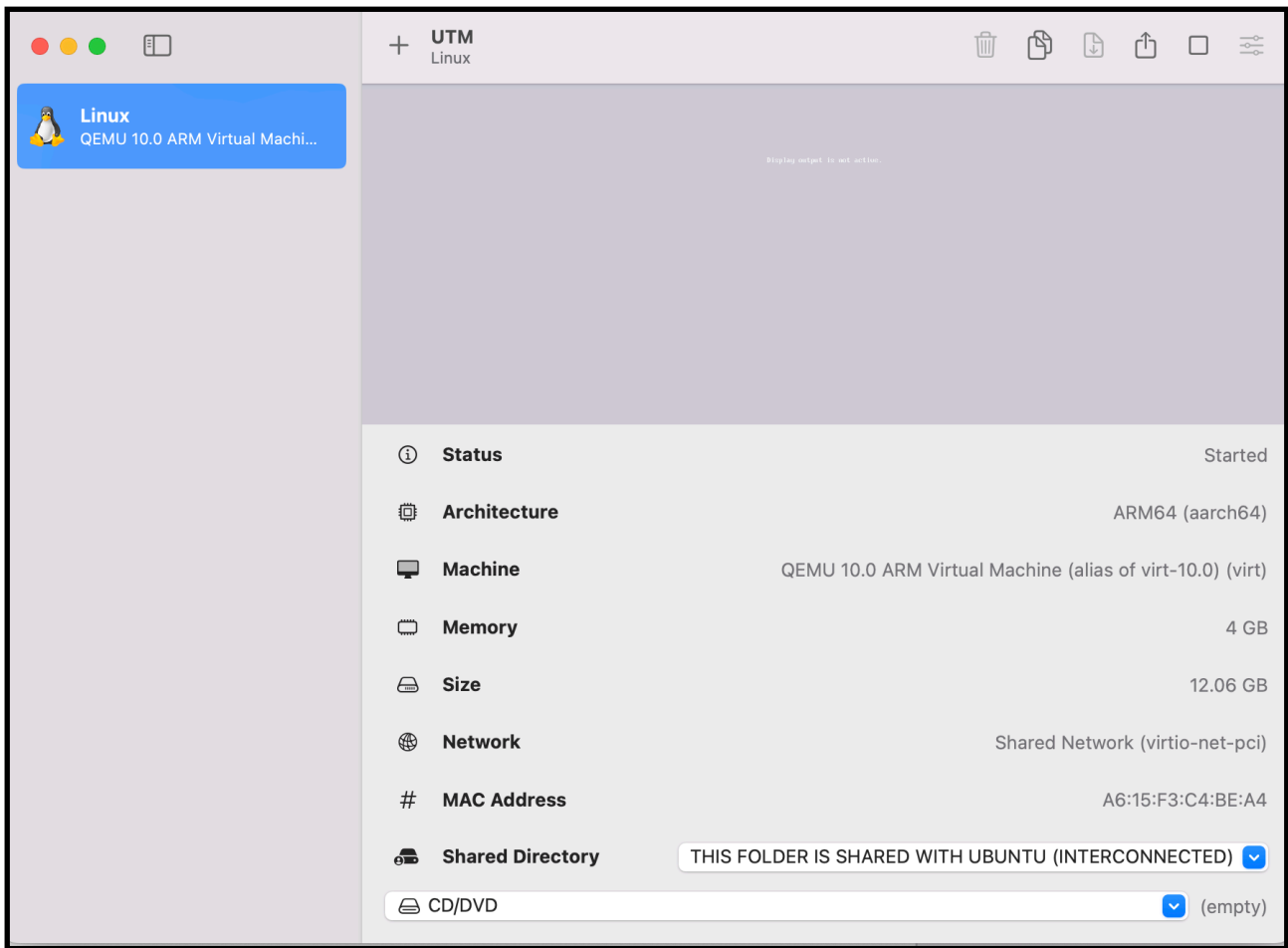
Steps to Create an Ubuntu Virtual Machine

Step 1: Install UTM on macOS (utm is available for Mac to create vm)

1. Download **UTM** for macOS.
2. Install and open UTM.
3. Click “+” → **Create** → **Virtualize** → **Linux**.

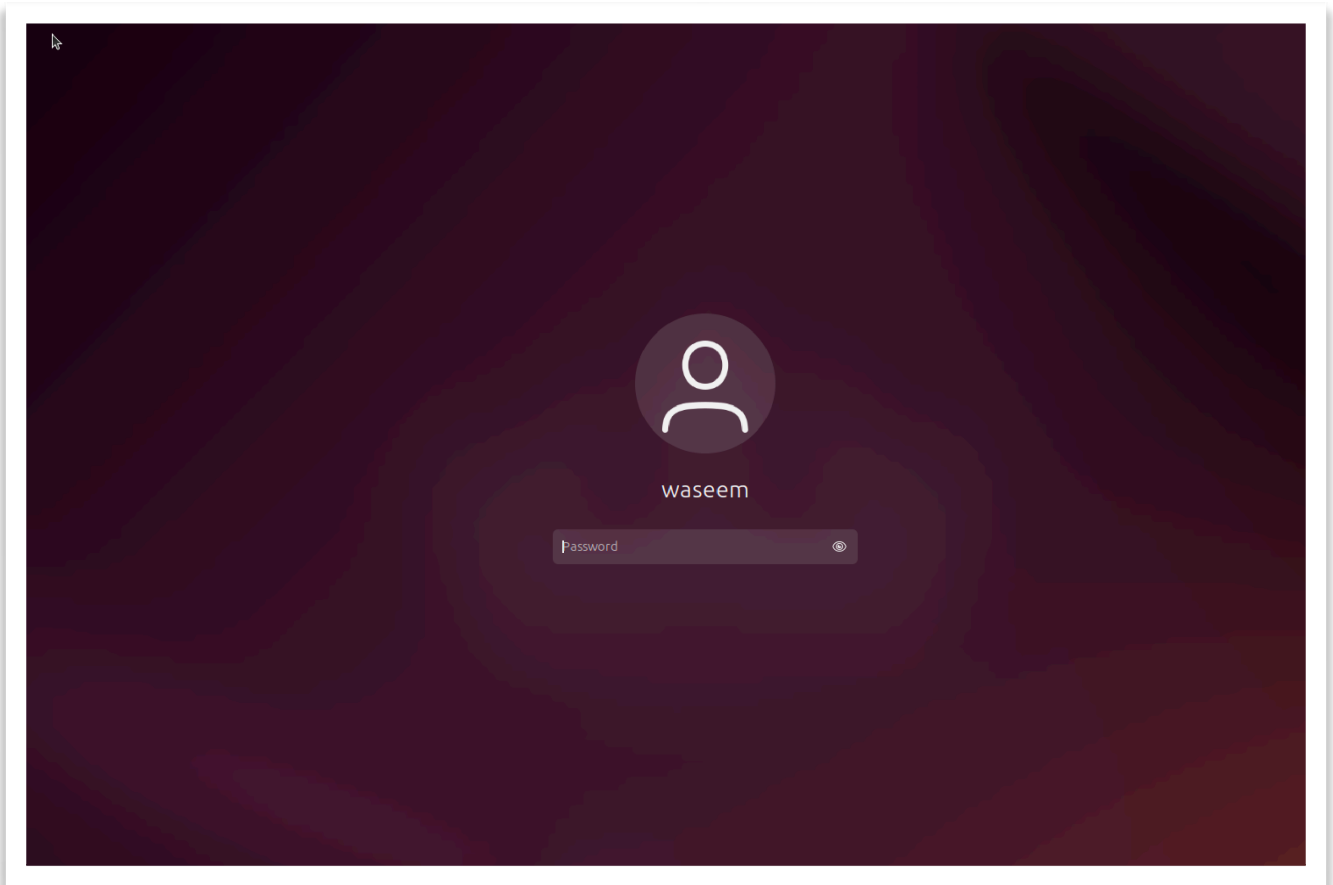
Step 2: Create Ubuntu Virtual Machine

1. Select **Ubuntu ISO image**.
2. Set VM details:
 - **Architecture:** ARM64 (aarch64)
 - **Memory:** 4 GB
 - **Storage:** ~12–20 GB
3. Choose **Shared Network (NAT)**.
4. Finish setup and start the VM.



Step 3: Install Ubuntu OS

1. VM boots from Ubuntu ISO.
2. Choose **Install Ubuntu**.
3. Select keyboard and region.
4. Create user account and password.
5. Complete installation and restart VM.



Step 4: Open Terminal and Switch to Root

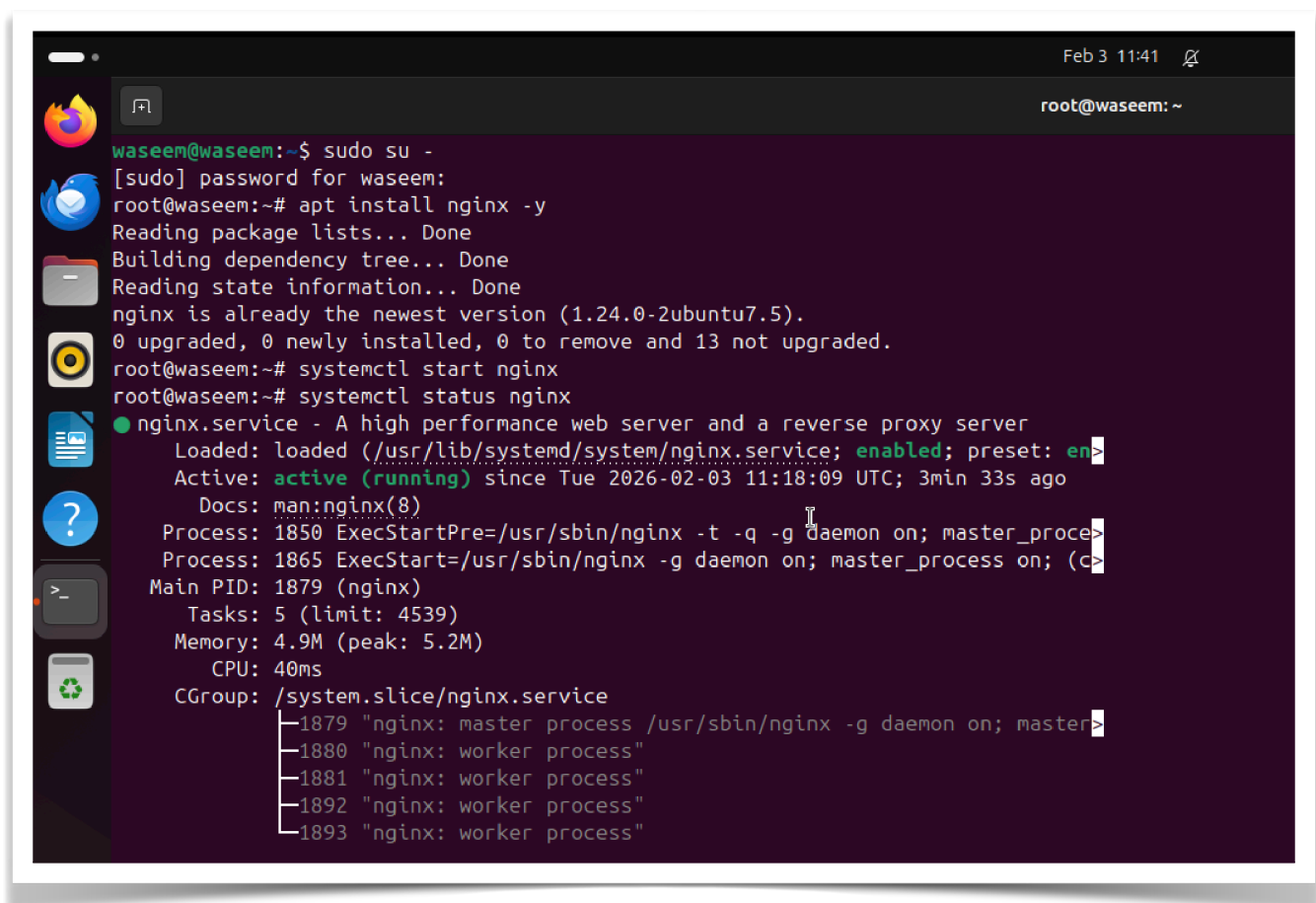
After login:

```
sudo su -
```

Step 5: Install Nginx Web Server

```
apt update
```

```
apt install nginx -y
```

A terminal window screenshot with a dark background. The window title bar shows 'Feb 3 11:41' and a close button. The terminal content shows a user switching to root and installing Nginx. The output indicates Nginx is already the latest version (1.24.0-2ubuntu7.5). The user then starts the service, and the status output shows it is active and running. The status details include the service name, loaded state, active state, documentation, process information, main PID, tasks, memory usage, CPU usage, and CGroup.

```
Feb 3 11:41
root@waseem: ~

waseem@waseem:~$ sudo su -
[sudo] password for waseem:
root@waseem:~# apt install nginx -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
nginx is already the newest version (1.24.0-2ubuntu7.5).
0 upgraded, 0 newly installed, 0 to remove and 13 not upgraded.
root@waseem:~# systemctl start nginx
root@waseem:~# systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: en>
   Active: active (running) since Tue 2026-02-03 11:18:09 UTC; 3min 33s ago
     Docs: man:nginx(8)
   Process: 1850 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_proce>
   Process: 1865 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (c>
  Main PID: 1879 (nginx)
    Tasks: 5 (limit: 4539)
   Memory: 4.9M (peak: 5.2M)
      CPU: 40ms
   CGroup: /system.slice/nginx.service
           └─1879 "nginx: master process /usr/sbin/nginx -g daemon on; master>
              └─1880 "nginx: worker process"
                 └─1881 "nginx: worker process"
                    └─1892 "nginx: worker process"
                       └─1893 "nginx: worker process"
```

Step 6: Start and Verify Nginx Service

```
systemctl start nginx
systemctl status nginx
```

Output shows:

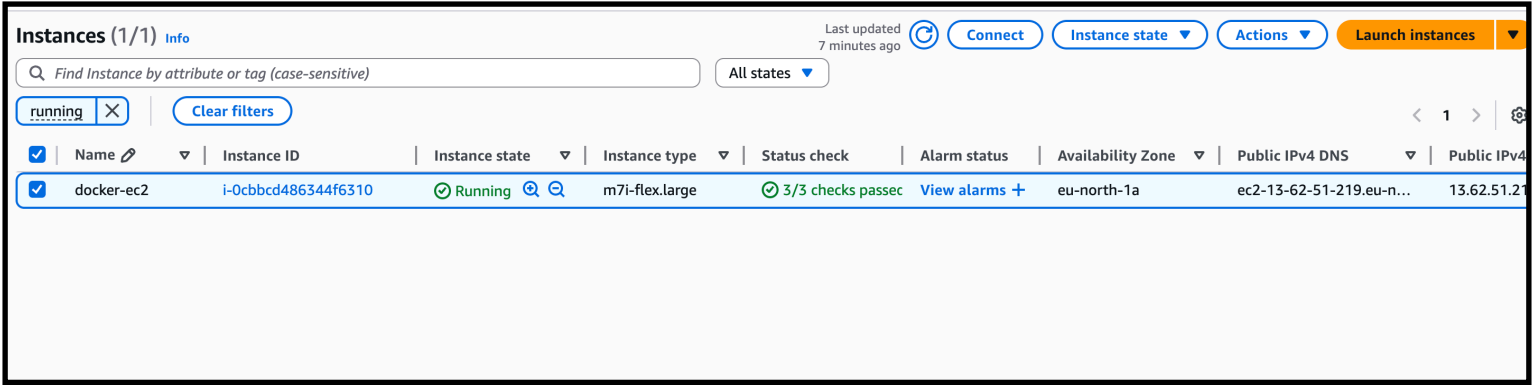
Active: active (running)
This confirms Nginx is running successfully.

Task :2

2). Install docker on ec2.

Prerequisites:

- 1.) ec2 with large volume
- 2.) t3.large



--> now run the server

Steps: inside server

--> yum update -y

--> yum install docker -y

--> systemctl start docker

--> systemctl status docker

```
Complete!  
[root@ip-172-31-20-146 ~]# yum update -y  
Last metadata expiration check: 0:00:12 ago on Wed Feb  4 06:33:49 2026.  
Dependencies resolved.  
Nothing to do.  
Complete!  
[root@ip-172-31-20-146 ~]# yum install docker -y  
Last metadata expiration check: 0:00:31 ago on Wed Feb  4 06:33:49 2026.  
Dependencies resolved.
```

Package	Architecture	Version
Installing: docker	x86_64	25.0.14-1.amzn2023.0.1
Installing dependencies:		
container-selinux	noarch	4:2.242.0-1.amzn2023.0.1
containerd	x86_64	2.1.5-1.amzn2023.0.1
iptables-libs	x86_64	1.8.8-3.amzn2023.0.1
iptables-nft	x86_64	1.8.8-3.amzn2023.0.1
libcgroup	x86_64	3.0-1.amzn2023.0.1
libnetfilter_conntrack	x86_64	1.0.8-2.amzn2023.0.1
libnftnl	x86_64	1.0.1-19.amzn2023.0.1
libnftnl	x86_64	1.2.2-2.amzn2023.0.1
pigz	x86_64	2.5-1.amzn2023.0.3
runc	x86_64	1.3.4-1.amzn2023.0.1

```
Transaction Summary  
-----  
Install 11 Packages  
  
Total download size: 74 M  
Installed size: 281 M  
Downloading Packages:  
(1/11): container-selinux-2.242.0-1.amzn2023.noarch.rpm  
(2/11): iptables-libs-1.8.8-3.amzn2023.0.2.x86_64.rpm  
(3/11): iptables-nft-1.8.8-3.amzn2023.0.2.x86_64.rpm  
(4/11): libcgroup-3.0-1.amzn2023.0.1.x86_64.rpm  
(5/11): libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64.rpm  
(6/11): libnftnl-1.0.1-19.amzn2023.0.2.x86_64.rpm  
(7/11): libnftnl-1.2.2-2.amzn2023.0.2.x86_64.rpm  
(8/11): pigz-2.5-1.amzn2023.0.3.x86_64.rpm  
(9/11): containerd-2.1.5-1.amzn2023.0.4.x86_64.rpm  
(10/11): runc-1.3.4-1.amzn2023.0.1.x86_64.rpm  
(11/11): docker-25.0.14-1.amzn2023.0.1.x86_64.rpm  
-----  
Total  
Running transaction check
```

```

complete.
[root@ip-172-31-20-146 ~]# systemctl start docker
[root@ip-172-31-20-146 ~]# systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; disabled; preset: disabled)
   Active: active (running) since Wed 2026-02-04 06:35:35 UTC; 7s ago
 TriggeredBy: ● docker.socket
      Docs: https://docs.docker.com
    Process: 27865 ExecStartPre=/bin/mkdir -p /run/docker (code=exited, status=0/SUCCESS)
    Process: 27866 ExecStartPre=/usr/libexec/docker/docker-setup-runtimes.sh (code=exited, status=0/SUCCESS)
   Main PID: 27867 (dockerd)
      Tasks: 9
     Memory: 29.5M
        CPU: 224ms
    CGroup: /system.slice/docker.service
            └─27867 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock --default-ulimit nofile=32768:

Feb 04 06:35:34 ip-172-31-20-146.eu-north-1.compute.internal systemd[1]: Starting docker.service - Docker Application Cont
Feb 04 06:35:34 ip-172-31-20-146.eu-north-1.compute.internal dockerd[27867]: time="2026-02-04T06:35:34.718025423Z" level=i
Feb 04 06:35:34 ip-172-31-20-146.eu-north-1.compute.internal dockerd[27867]: time="2026-02-04T06:35:34.773972194Z" level=i
Feb 04 06:35:35 ip-172-31-20-146.eu-north-1.compute.internal dockerd[27867]: time="2026-02-04T06:35:35.141909801Z" level=i
Feb 04 06:35:35 ip-172-31-20-146.eu-north-1.compute.internal dockerd[27867]: time="2026-02-04T06:35:35.158765368Z" level=i
Feb 04 06:35:35 ip-172-31-20-146.eu-north-1.compute.internal dockerd[27867]: time="2026-02-04T06:35:35.158859993Z" level=i
Feb 04 06:35:35 ip-172-31-20-146.eu-north-1.compute.internal dockerd[27867]: time="2026-02-04T06:35:35.189169291Z" level=i
Feb 04 06:35:35 ip-172-31-20-146.eu-north-1.compute.internal systemd[1]: Started docker.service - Docker Application Conta
[root@ip-172-31-20-146 ~]# sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; disabled; preset: disabled)
   Active: active (running) since Wed 2026-02-04 06:35:35 UTC; 23s ago
 TriggeredBy: ● docker.socket
      Docs: https://docs.docker.com
    Process: 27865 ExecStartPre=/bin/mkdir -p /run/docker (code=exited, status=0/SUCCESS)
    Process: 27866 ExecStartPre=/usr/libexec/docker/docker-setup-runtimes.sh (code=exited, status=0/SUCCESS)
   Main PID: 27867 (dockerd)
      Tasks: 9
     Memory: 29.5M
        CPU: 225ms
    CGroup: /system.slice/docker.service
            └─27867 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock --default-ulimit nofile=32768:

Feb 04 06:35:34 ip-172-31-20-146.eu-north-1.compute.internal systemd[1]: Starting docker.service - Docker Application Cont
Feb 04 06:35:34 ip-172-31-20-146.eu-north-1.compute.internal dockerd[27867]: time="2026-02-04T06:35:34.718025423Z" level=i
Feb 04 06:35:34 ip-172-31-20-146.eu-north-1.compute.internal dockerd[27867]: time="2026-02-04T06:35:34.773972194Z" level=i
Feb 04 06:35:35 ip-172-31-20-146.eu-north-1.compute.internal dockerd[27867]: time="2026-02-04T06:35:35.141909801Z" level=i
Feb 04 06:35:35 ip-172-31-20-146.eu-north-1.compute.internal dockerd[27867]: time="2026-02-04T06:35:35.158765368Z" level=i
Feb 04 06:35:35 ip-172-31-20-146.eu-north-1.compute.internal dockerd[27867]: time="2026-02-04T06:35:35.158859993Z" level=i
Feb 04 06:35:35 ip-172-31-20-146.eu-north-1.compute.internal dockerd[27867]: time="2026-02-04T06:35:35.189169291Z" level=i
Feb 04 06:35:35 ip-172-31-20-146.eu-north-1.compute.internal systemd[1]: Started docker.service - Docker Application Conta
[root@ip-172-31-20-146 ~]#

```

—> docker is now running and active

Task :3

3) Pull 5 docker images.(nginx,apache tomcat,ubuntu,jenkins,sonarqube)

--> docker pull nginx

```
[root@ip-172-31-20-146 ~]# docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
0c8d55a45c0d: Pull complete
173e7a5d3717: Pull complete
2711d25abbb0: Pull complete
359cde133485: Pull complete
acc398fdf80e: Pull complete
bc4d011570c3: Pull complete
1eb69ddd819c: Pull complete
Digest: sha256:9dd288848f4495869f76676e419ae2d767ca99fece2ec37ec0261f9fdaab5204
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
[root@ip-172-31-20-146 ~]# docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
nginx          latest    248d2326f351   28 hours ago   161MB
[root@ip-172-31-20-146 ~]#
```

--> docker pull tomcat

--> to check the images -> docker images

```
[root@ip-172-31-20-146 ~]# docker pull tomcat
Using default tag: latest
latest: Pulling from library/tomcat
a3629ac5b9f4: Pull complete
bd41d65c3828: Pull complete
8be760c2a9a8: Pull complete
4f4fb700ef54: Pull complete
d0702afe109d: Pull complete
d99c7608407b: Pull complete
2ca1458a9031: Pull complete
Digest: sha256:6af1184ab272d631d62a712260ac858010a898b835c088cbe447653cdc1d3d97
Status: Downloaded newer image for tomcat:latest
docker.io/library/tomcat:latest
[root@ip-172-31-20-146 ~]# docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
nginx          latest    248d2326f351   29 hours ago   161MB
tomcat         latest    fe1a24e4a16c   8 days ago     412MB
[root@ip-172-31-20-146 ~]#
```

--> docker pull ubuntu

```
tomcat latest fe1a24e4a16c 8 days ago 412MB
[root@ip-172-31-20-146 ~]# docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
a3629ac5b9f4: Already exists
Digest: sha256:cd1dba651b3080c3686ecf4e3c4220f026b521fb76978881737d24f200828b2b
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest
[root@ip-172-31-20-146 ~]# docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
nginx         latest    248d2326f351   29 hours ago   161MB
tomcat        latest    fe1a24e4a16c   8 days ago     412MB
ubuntu        latest    493218ed0f40   3 weeks ago    78.1MB
[root@ip-172-31-20-146 ~]#
```

--> docker pull jenkins

Why docker pull jenkins/jenkins

and NOT docker pull jenkins?

1 Docker image naming rule

Docker image name format is:

`<namespace>/<repository>:<tag>`

- **namespace** → who owns the image
- **repository** → image name
- **tag** → version (default = latest)

2 What happens when you run:

`docker pull jenkins`

Docker assumes:

`docker.io/library/jenkins`

👉 Problem:

There is **NO official jenkins image** under `library/`.

So Docker says:

`pull access denied, repository does not exist`

```
ubuntu:latest 493218ed0f40 3 weeks ago 78.1MB
[root@ip-172-31-20-146 ~]# docker pull jenkins
Using default tag: latest
Error response from daemon: manifest for jenkins:latest not found: manifest unknown: manifest unknown
[root@ip-172-31-20-146 ~]#
```

3 Why `jenkins/jenkins` works

`docker pull jenkins/jenkins`

Means:

`docker.io/jenkins/jenkins`

- **jenkins** (namespace) → Jenkins organization
- **jenkins** (repo) → Jenkins image

✓ This is the **official image published by the Jenkins team**

That's why it works

—-> `docker pull jenkins/jenkins`


```
Error response from daemon: manifest for jenkins/latest not found: manifest unknown: manifest unknown
[root@ip-172-31-20-146 ~]# docker pull jenkins/jenkins
Using default tag: latest
latest: Pulling from jenkins/jenkins
53c88f1dfeb7: Pull complete
7006c93f4ecd: Pull complete
0d5e30284279: Pull complete
aea1c7bd2c60: Pull complete
9904a9c50dc1: Pull complete
e7d0072c9676: Pull complete
4290eef74c05: Pull complete
33e31423140b: Pull complete
f504968eed54: Pull complete
ff9fdc9d0ffb: Pull complete
4fe16e51cec6: Pull complete
b650ec4dbb1c: Pull complete
Digest: sha256:e9df903e1d9a22132cd26463cfa8cdaa70f53d0dc0834e31f396de687294af22
Status: Downloaded newer image for jenkins/jenkins:latest
docker.io/jenkins/jenkins:latest
[root@ip-172-31-20-146 ~]# docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
jenkins/jenkins     latest             64cb03009c23       15 hours ago       487MB
nginx               latest             248d2326f351       29 hours ago       161MB
tomcat              latest             fe1a24e4a16c       8 days ago         412MB
ubuntu              latest             493218ed0f40       3 weeks ago        78.1MB
[root@ip-172-31-20-146 ~]#
```



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jenkins/jenkins

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IMAGE

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INTEGRATION & DELIVERY

MONITORING & OBSERVABILITY

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Tags

—> docker pull sonarqube

```
[root@ip-172-31-20-146 ~]# docker pull sonarqube
Using default tag: latest
latest: Pulling from library/sonarqube
a3629ac5b9f4: Already exists
220e5f46c255: Pull complete
a3307ab29355: Pull complete
24ebe7eabaf2: Pull complete
df8fd45c97cd: Pull complete
30d803a36141: Pull complete
b499eabd66bb: Pull complete
4f4fb700ef54: Pull complete
Digest: sha256:1b1117276ca3ae9a692b1bd9b75cc28289de958675f217079f8d0a776bef64d8
Status: Downloaded newer image for sonarqube:latest
docker.io/library/sonarqube:latest
[root@ip-172-31-20-146 ~]# docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
jenkins/jenkins	latest	64cb03009c23	15 hours ago	487MB
nginx	latest	248d2326f351	29 hours ago	161MB
sonarqube	latest	aec8336830ce	5 days ago	1.45GB
tomcat	latest	fe1a24e4a16c	8 days ago	412MB
ubuntu	latest	493218ed0f40	3 weeks ago	78.1MB

```
[root@ip-172-31-20-146 ~]#
```

—> pulled all 5 images

Task 4:

4) Run nginx container and expose on port 81.

Command to expose nginx on port 81:

```
docker run -d -p 81:80 nginx
```

```
ubuntu          latest         493218ed0f40   3 weeks ago     78.1MB
[root@ip-172-31-20-146 ~]# docker run -d -p 81:80 nginx
c6c8c08336c32973539cd29d32ca69edb53fd54a7beae6bd04f294e4436abc80
[root@ip-172-31-20-146 ~]# docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
c6c8c08336c3	nginx	"/docker-entrypoint..."	12 seconds ago	Up 11 seconds	0.0.0.0:81->80/tcp, :::81->80/tcp	mystifying_bhabha

```
[root@ip-172-31-20-146 ~]# docker run -d \
```

run

- Creates a **new container**
- Starts it immediately

👉 If the image is not present, Docker **downloads it first**.

◆ **-d**

Detached mode

- Runs the container in the **background**
- Your terminal stays free

Without **-d**, nginx logs would block your terminal.

◆ **-p 81:80**

Port mapping (most important part)

Format:

`-host-port : container-port`

So:

- **81** → your machine / server port
- **80** → nginx port **inside the container**

Meaning:

When someone accesses **port 81** on the host, traffic is sent to **port 80** inside the nginx container.

What happens after running this command?

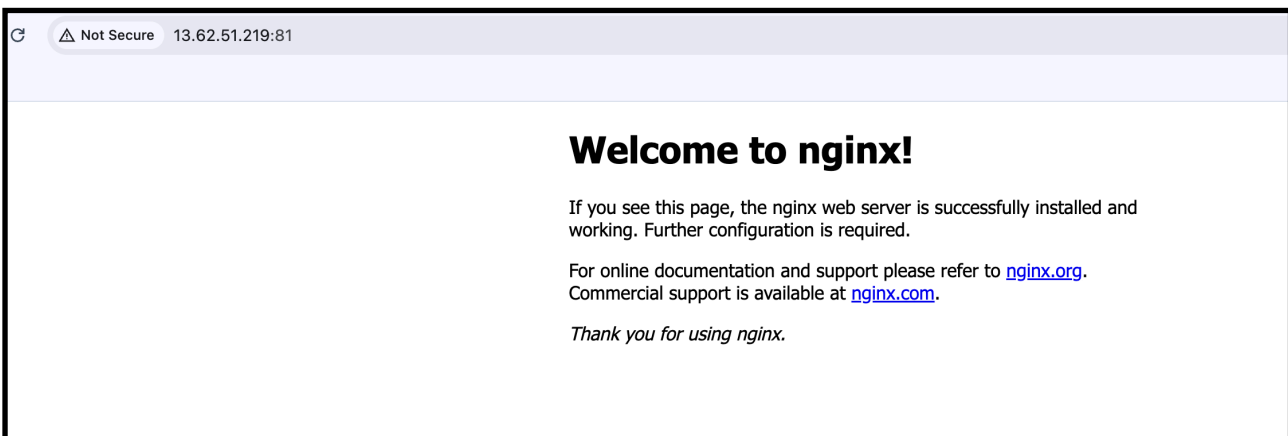
- nginx runs inside a container
- nginx listens on **port 80** internally
- Docker exposes it on **port 81** externally

Verification:

`docker ps`

```
See 'docker run --help'.
[root@ip-172-31-20-146 ~]# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
c6c8c08336c3   nginx     "/docker-entrypoint..." About a minute Up About a minute 0.0.0.0:81->80/tcp, :::81->80/tcp   mystifying_bhabha
[root@ip-172-31-20-146 ~]#
```

—-> check your public ip:81



Task:5

5).Delete the apache tomcat image from local

Command

`docker rmi tomcat`

—> What this command does

- `docker rmi` → remove image
- `tomcat` → apache tomcat image name

```
[root@ip-172-31-20-146 ~]# docker images
REPOSITORY      TAG       IMAGE ID       CREATED        SIZE
jenkins/jenkins  latest    64cb03009c23   15 hours ago   487MB
nginx            latest    248d2326f351   29 hours ago   161MB
sonarqube        latest    aec8336830ce    5 days ago     1.45GB
tomcat           latest    fe1a24e4a16c    8 days ago     412MB
ubuntu           latest    493218ed0f40    3 weeks ago    78.1MB
[root@ip-172-31-20-146 ~]# docker rmv tomcat
docker: 'rmv' is not a docker command.
See 'docker --help'
[root@ip-172-31-20-146 ~]# docker rmi tomcat
Untagged: tomcat:latest
Deleted: sha256:6af1184ab272d631d62a712260ac858010a898b835c088cbe447653cdc1d3d97
Deleted: sha256:fe1a24e4a16c9df595f3b8afe54a1c2a6d23ccf83be09b919bc2fb91c6390b56
Deleted: sha256:a35aebfcb4887b720b215be1489093672b847c3f579d82298ac910963d8f9093
Deleted: sha256:0cfce22b3fc26b07db7c6909372470ba7461cb3a027337cb4161746e7293b3d6
Deleted: sha256:1c146fd53edfd8e7597a4b378c659c5f46283b2228c884c348001b439f302641
Deleted: sha256:55a7dbb16d051b26c139fd2bf0764a805574b332ba2ac0a74df3960a287ced84
Deleted: sha256:7bdeec6c49307b38e6405f199f00b6b0e91a1c3f2189e7cfb5ce4608c91731b4
Deleted: sha256:229685cb26f3d1db706ac542eee7b7c68ae21e392569b47de3dd7f57e2791437
Deleted: sha256:b29476111056abd42d1f0fd174414af44bceb8632d58e619f12ada13193a89b5
Deleted: sha256:4aba6fa6825957086b4eb801afd7c072e057d1628884457090f6c2d519020e31
[root@ip-172-31-20-146 ~]#
```

This deletes the **Tomcat** image from your local Docker host.

—> now check you docker images

```
Deleted: sha256:4aba6fa6825957086b4eb801afd7c072e057d1628884457090f6c2d519020e31
[root@ip-172-31-20-146 ~]# docker images
REPOSITORY      TAG       IMAGE ID       CREATED        SIZE
jenkins/jenkins  latest    64cb03009c23   15 hours ago   487MB
nginx            latest    248d2326f351   29 hours ago   161MB
sonarqube        latest    aec8336830ce    5 days ago     1.45GB
ubuntu           latest    493218ed0f40    3 weeks ago    78.1MB
[root@ip-172-31-20-146 ~]#
```

Task:6

6).Inspect the jenkins image,sonarqube image.

Command for inspect the Jenkins image and sonarqube is

—> docker inspect jenkins/jenkins

```
[root@ip-172-31-20-146 ~]# docker inspect jenkins/jenkins
[
  {
    "Id": "sha256:64cb03009c233c50be439740447bf71d4b5b1531888f5112ee7f4cd91737f4f3",
    "RepoTags": [
      "jenkins/jenkins:latest"
    ],
    "RepoDigests": [
      "jenkins/jenkins@sha256:e9df903e1d9a22132cd26463cfa8cdaa70f53d0dc0834e31f396de687294af22"
    ],
    "Parent": "",
    "Comment": "buildkit.dockerfile.v0",
    "Created": "2026-02-03T16:06:06.357599106Z",
    "Container": "",
    "ContainerConfig": {
      "Hostname": "",
      "Domainname": "",
      "User": "",
      "AttachStdin": false,
      "AttachStdout": false,
      "AttachStderr": false,
      "Tty": false,
      "OpenStdin": false,
      "StdinOnce": false,
      "Env": null,
      "Cmd": null,
      "Image": "",
      "Volumes": null,
      "WorkingDir": "",
      "Entrypoint": null,
      "OnBuild": null,
      "Labels": null
    },
    "DockerVersion": "",
    "Author": "",
    "Config": {
      "Hostname": "",
      "Domainname": "",
      "User": "jenkins",
      "AttachStdin": false,
      "AttachStdout": false,
      "AttachStderr": false,
      "ExposedPorts": {
        "50000/tcp": {},
        "8080/tcp": {}
      },
      "Tty": false,
      "OpenStdin": false,
      "StdinOnce": false,
      "Env": [
        "PATH=/opt/java/openjdk/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
        "LANG=C.UTF-8",
        "JENKINS_HOME=/var/jenkins_home",
        "JENKINS_SLAVE_AGENT_PORT=50000",
        "REF=/usr/share/jenkins/ref",
        "JENKINS_UC=https://updates.jenkins.io",
        "JENKINS_UC_EXPERIMENTAL=https://updates.jenkins.io/experimental",
        "JENKINS_INCREMENTALS_REPO_MIRROR=https://repo.jenkins-ci.org/incrementals",
        "COPY_REFERENCE_FILE_LOG=/var/jenkins_home/copy_reference_file.log",
        "JAVA_HOME=/opt/java/openjdk"
      ]
    },
    "Cmd": null,
  }
]
```

--> docker inspect sonarqube

```
[root@ip-172-31-20-146 ~]# docker inspect sonarqube
[
  {
    "Id": "sha256:aec8336830cedd2c0cb9161fcf3f577b2f0bc0d4b46324d82814385d927dee06",
    "RepoTags": [
      "sonarqube:latest"
    ],
    "RepoDigests": [
      "sonarqube@sha256:1b1117276ca3ae9a692b1bd9b75cc28289de958675f217079f8d0a776bef64d8"
    ],
    "Parent": "",
    "Comment": "buildkit.dockerfile.v0",
    "Created": "2026-01-29T21:36:33.672046095Z",
    "Container": "",
    "ContainerConfig": {
      "Hostname": "",
      "Domainname": "",
      "User": "",
      "AttachStdin": false,
      "AttachStdout": false,
      "AttachStderr": false,
      "Tty": false,
      "OpenStdin": false,
      "StdinOnce": false,
      "Env": null,
      "Cmd": null,
      "Image": "",
      "Volumes": null,
      "WorkingDir": "",
      "Entrypoint": null,
      "OnBuild": null,
      "Labels": null
    },
    "DockerVersion": "",
    "Author": "",
    "Config": {
      "Hostname": "",
      "Domainname": "",
      "User": "sonarqube",
      "AttachStdin": false,
      "AttachStdout": false,
      "AttachStderr": false,
      "ExposedPorts": {
        "9000/tcp": {}
      },
      "Tty": false,
      "OpenStdin": false,
      "StdinOnce": false,
      "Env": [
        "PATH=/opt/java/openjdk/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
        "JAVA_HOME=/opt/java/openjdk",
        "LANG=en_US.UTF-8",
        "LANGUAGE=en_US:en",
        "LC_ALL=en_US.UTF-8",
        "JAVA_VERSION=jdk-21.0.9+10",
        "DOCKER_RUNNING=true",
        "SONARQUBE_HOME=/opt/sonarqube",
        "SONAR_VERSION=26.1.0.118079",

```

What docker image inspect does

It shows **detailed information** about the image, like:

- Image ID
- Created date
- OS & architecture
- Environment variables

- Exposed ports
- Entrypoint & CMD
- Layers & size

Before inspect, you can check:

`docker images`

Task:7

7) Run jenkins container and run one sample job.

command

`docker run -d -p 8080:8080 -p 50000:50000 jenkins/jenkins`

```
[root@ip-172-31-20-146 ~]# docker run -d -p 8080:8080 -p 50000:50000 jenkins/jenkins
ac110567ca1a33dad651cbb89ebb932989f1f4f0f2bf6a8ec4a7d0f7c6cf652
[[root@ip-172-31-20-146 ~]# docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS
--------------	-------	---------	---------	--------

Check if Jenkins running

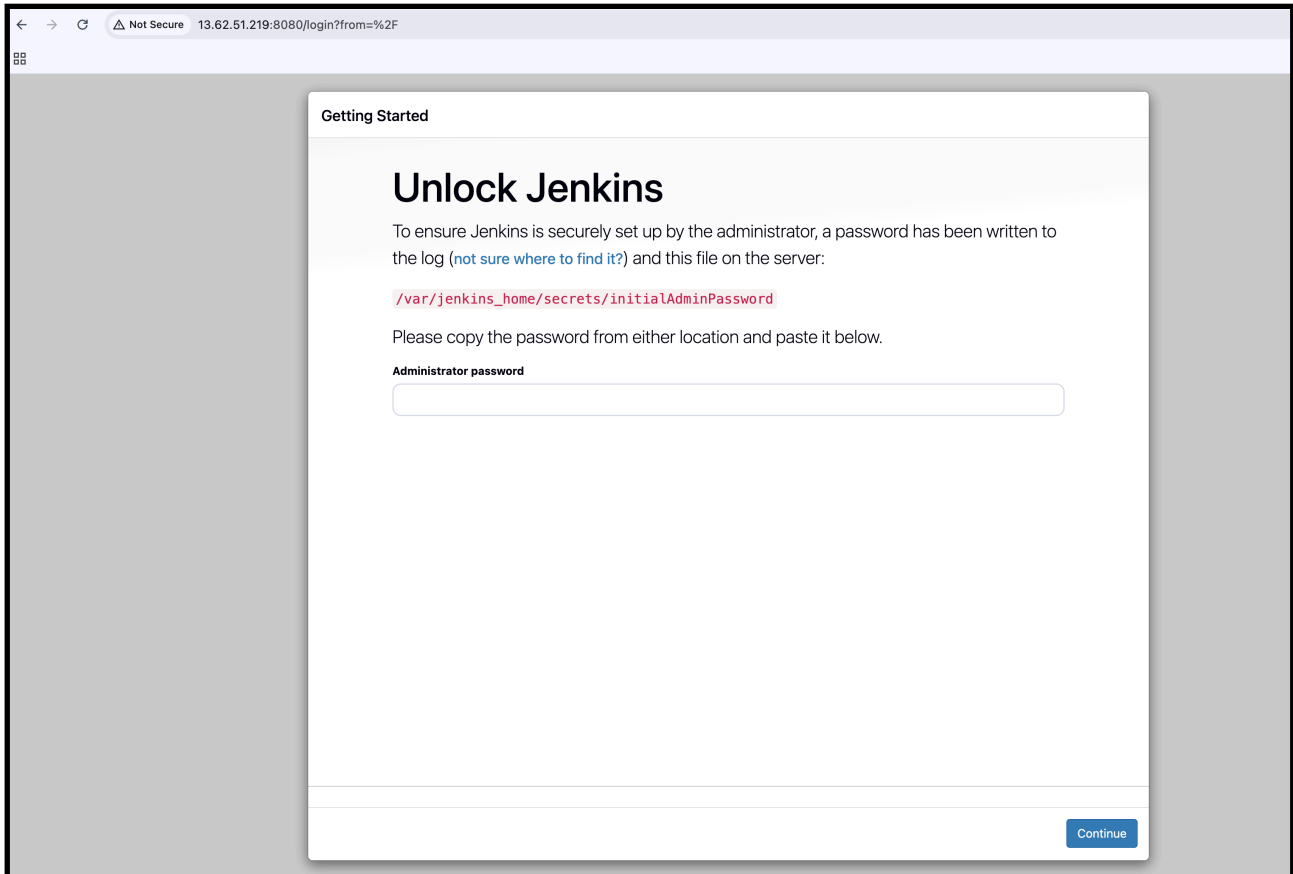
—> Docker ps

```
[root@ip-172-31-20-146 ~]# docker run -d -p 8080:8080 -p 50000:50000 jenkins/jenkins
ac110567ca1a33dad651cbb89ebb932989f1f4f0f2bf6a8ec4a7d0f7c6cf652
[[root@ip-172-31-20-146 ~]# docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAME
ac110567ca1a	jenkins/jenkins	"/usr/bin/tini -- /u..."	14 seconds ago	Up 14 seconds	0.0.0.0:8080->8080/tcp, ::8080->8080/tcp, 0.0.0.0:50000->50000/tcp, ::50000->50000/tcp	penstemon
c6c8c08336c3	nginx	"/docker-entrypoint..."	22 minutes ago	Up 22 minutes	0.0.0.0:81->80/tcp, ::81->80/tcp	mystic

```
[root@ip-172-31-20-146 ~]#
```


—> now go and check the public ip:8080



Get initial admin password

Run:

```
docker exec pensive_mcclintock cat /var/jenkins_home/
secrets/initialAdminPassword
```

—> pensive_mcclintock is the name of Jenkins

```
.0:50000->50000/tcp, :::50000->50000/tcp
NAMES
pensive_mcclintock
mystifying_bhabha
```

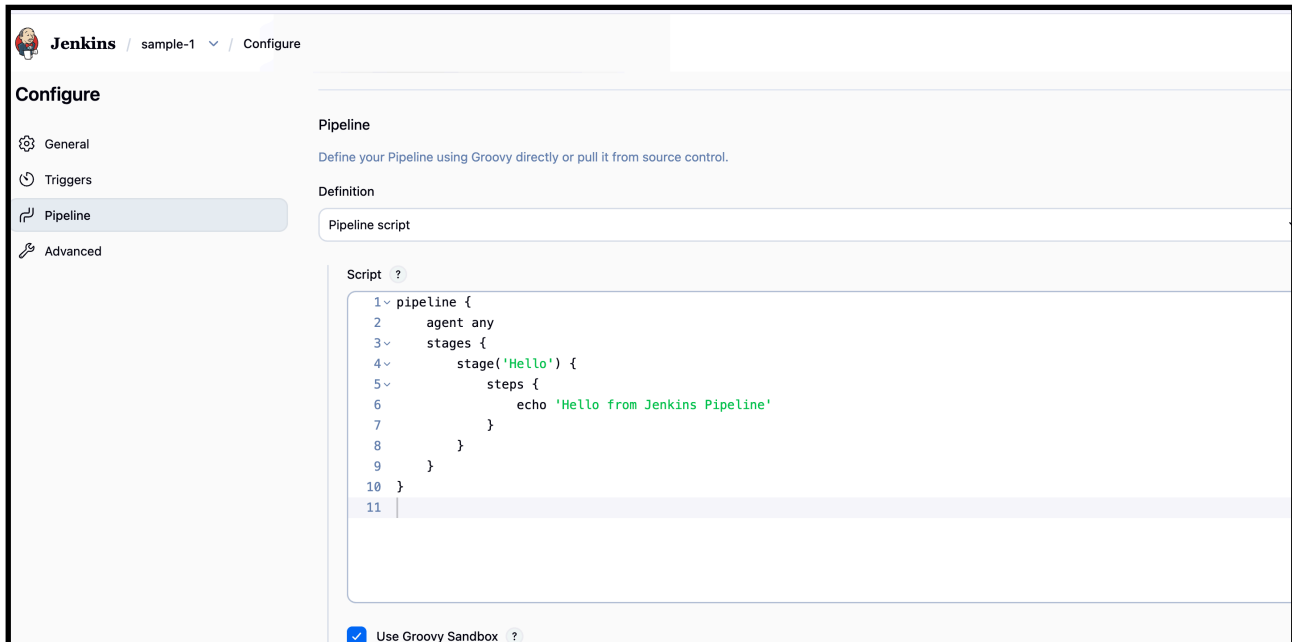
```
[root@ip-172-31-20-146 ~]# docker exec pensive_mcclintock cat /var/jenkins_home/secrets/initialAdminPassword  
bd178b91dfd047a2a704dc3f83a010cb
```

—> now copy the password and paste in Jenkins gui

—-> and after login

—-> create a job and choose pipeline

—-> add the pipeline



—-> now run the job



Status

✓ sample-1

</> Changes

▶ Build Now

⚙️ Configure

🗑️ Delete Pipeline

🔍 Full Stage View

📄 Stages

✎️ Rename

❓ Pipeline Syntax

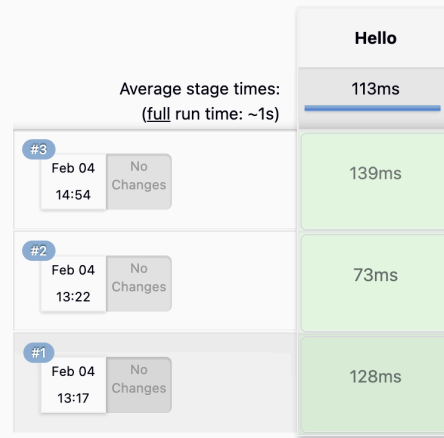
Builds >

🔍 Filter

Today

✓ #3 9:24 AM

Stage View



Permalinks